Talk 2

Tuesday, March 24, 2026 1:28 PM

Review modular categories:

let k be an algebraically closed field

Def: A fusion category is a finite k-linear semisimple abelian tensor category $\mathcal{C}$. meaning:

- we have a tensor product
  $\mathcal{C} \otimes \mathcal{C} \to \mathcal{C}$ (bifunctor)
  coherent it satisfies the pentagon axiom with a natural iso
  $\alpha_{X,Y,Z} : (X \otimes Y) \otimes Z \to X \otimes (Y \otimes Z)$ and $l_X : 1 \otimes X \to X$ $r_X : X \otimes 1 \to X$
- $\mathrm{Hom}_{\mathcal{C}}(X,Y)$ has str of k-module with bilinear composition (k-linear
- 0 object exists, direct sums exist $X \oplus Y$ with kernel and cokernel (abelian
- $V \cong \bigoplus_{i \in I} N_i V_i$ (semisimple) $\forall V \in \mathrm{obj}(\mathcal{C})$
- $\mathrm{Hom}_{\mathcal{C}}(X,Y)$ is finite and we have finitly many simple objects.

Rmk:

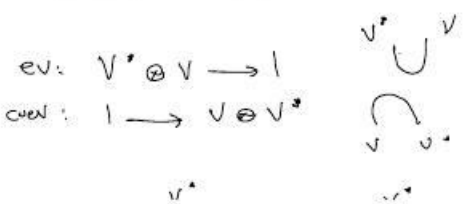
- $V_i \otimes V_j = \bigoplus N_{ij}^{k} V_k$ ← Fusion rules
- let $f: X \to Y$
  $\ker(f) \to X \xrightarrow{f} Y \to \mathrm{coker}(f)$ $\Rightarrow \mathrm{End}(V_i) \cong \mathbb{C}$
  $\ker(f) = 0$, $\mathrm{coker}(f) = 0$

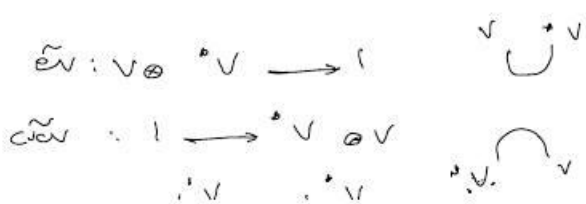
Graphical description: $f: X \to Y$

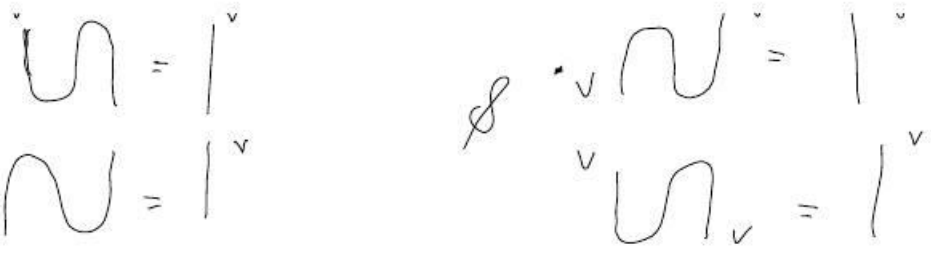


Rigidity:

Def: $\mathcal{C}$ is called left/right rigid if $\exists$ a left/right dual of V ${}^*V$, V^* s.t





s.t 

it's rigid if $\forall V \in obj(C)$ V has left and right duals!

Rmk: this allows the definition of right and left monoidal functors

$$?^* : C \longrightarrow C^{op}$$

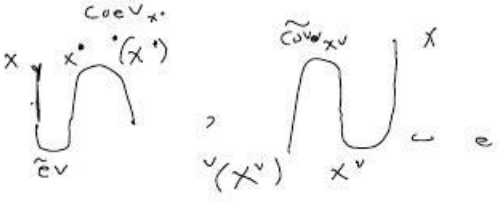
$$^*? : C \longrightarrow C^{op}$$

Then for a rigid category we have

$$(^*?)^{op} \circ (?^*) : C \longrightarrow C$$

which induces the following iso

$$^*(X^*) \simeq X \simeq (^*X)^*$$



Def: A pivotal (sovereign) category is a rigid category s.t the i left and right functors coincide as monoidal categories

That is $\forall x \in obj(C)$

$\{ x^*, ev, \tilde{ev} \}$ coevs are just the opposite

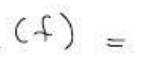
$ev, \tilde{ev}$ are non degenerate!

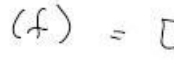
Def: the double dual functor in a pivital category

$$?^{**} : C \longrightarrow C$$

induces $X \simeq X^{**}$ (monidal natural isomorphism)

Def: let C be pivotal $f : X \to X$

$$Tr_L(f) = \text{  } = ev_X (id_{X^*} \otimes f) \circ \overline{coev}_X$$

$$Tr_r(f) = \text{  } = \tilde{ev}_X (f \otimes id_{X^*}) \circ coev_X$$

Def: A spherical category is a pivtal category s.t

$$\mathrm{tr}_L(f) = \mathrm{tr}_R(f)$$

so that we have a good notion of trace!

Braiding: A braiding str on a category $\mathcal{C}$ consisting of natural isom $c_{X,Y}: X \otimes Y \longrightarrow Y \otimes X$ satisfying hexagon axioms.



Def: the braiding is called non degenerate if $\mathcal{Z}_2(\mathcal{C})$ modular center has only $\mathbb{1}$ as transparent object!

Rmk: if a fusion category doesnot admit a braiding then just consider it's center $\mathcal{Z}(\mathcal{C})$.

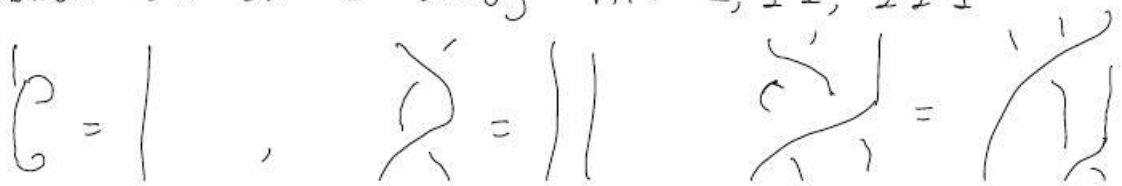
Ribbon Structure: let $\mathcal{C}$ be a braided category $\mathcal{C}$ has a twist if there is a natural isomorphism θ of the identity functor $id_{\mathcal{C}}$ s.t

$$\theta_{X\otimes Y} = (\theta_X \otimes \theta_Y) \circ c_{Y,X} \circ c_{X,Y}$$

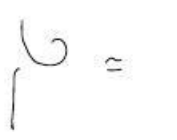
the twist is called ribbon if $(\theta_X)^* = \theta_X$.



prop: ribbon str on $\hat{\mathcal{C}}$ (fusion) satisfy RM I, II, III



Prop: The ribbon str on $\mathcal{C}$ a fusion category is equivalent to a spherical pivotal structure

basically  left and right twist coincide.

one can show that

$$(\theta^r_x)^* = (\theta^L_{x^*}) \quad , \quad (\theta^L_x)^* = \theta^r_{x^*}$$] in a braided pivotal categ

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Def: let $\mathcal{C}, \mathcal{D}$ be two ribbon fusion categories $\mathcal{C}, \mathcal{D}$ are ribb equivalent if $\exists$ a braided monoidal functor F compatible with associat braiding and twist.

Prop: A ribbon equivalence induces a permutation on $\mathrm{Irr}(\mathcal{C})$.

Def: A modular tensor categor $\mathcal{C}$ is a nondegenerate ribbon fusion category!

Invariants of Links:

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we can then assign numerical invariants to links

let L be a Markov closure of

$\hat{b}$ of an n-strand $b \in B_n$ ← braid group

let $x_i \in \mathcal{C}$ be labels of L. then the invar assoiated to L is

$$L_{x_1, x_2, \ldots, x_n} = \mathrm{Tr}(b) =$$ 

Def: we define the quantum dimension

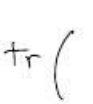
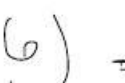
$\dim(x) =$  $\forall x \in \mathrm{Irr}(\mathcal{C})$

$d_x = tr(id_x) =$

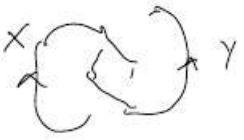
global dimension $\mathcal{D}^2 = \sum d_x^2$

- Since $\theta_x : X \longrightarrow X$ twist $\in End(X)$ if X is simple

then 

$$\Rightarrow \quad \theta_x = \frac{1}{d_x} tr\left(\begin{array}{c} \\ | \end{array}\right) = \frac{1}{d_x}$$  $$= \frac{1}{d_x}$$ 

we can also do this for the hopf link

$$S_{x,y} = Tr\left(C_{y^*,x}\, C_{x,y^*}\right) =$$ 

S- matrix

$$T_{x,y} = \theta_x \delta_{x,y} \quad \leftarrow \quad T \text{ matrix}$$

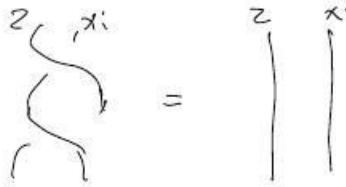
Prop: T and S give a representation of the modular group $SL(2,\mathbb{Z})$.

Prop: In a ribbon fusion category,

if S matrix is invertible $\Rightarrow$ $\mathcal{C}$ is non degenerate

proof: suppose $\mathcal{C}$ is degenerate

$\Rightarrow \exists\, Z \in \mathcal{C}$ that is transparent



$\forall \quad x_i \in Irr(\mathcal{C})$

Z can be chosen to simple since $\mathcal{C}$ is semisimple

then $S_{Zx_i} = d_Z d_{x_i}$ which is identical to

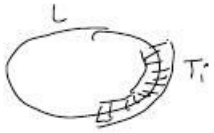
$$= S_{1 x_i}$$

$\Rightarrow$ S is not invertible!

Rmk: other direction is a bit involved!

Review topological facts

Def: let L be a link in S^3.



T_i tubular nbhd around L

let T_0 be a standard fixed Torus



suppose we have $f_i : \partial T_i \xrightarrow{\sim} \partial T_0$

Then we can define a new manifold

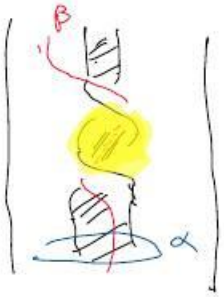
$$M_{L,f} = \left(S^3 \setminus \left(\cup_i T_i^{int}\right)\right) \cup_{f_i} (T_0)^N \quad \leftarrow \text{number of components of } L$$

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we can further fix what f_i does!

This can be done if L is framed

The framing can be thought as an arrow that moves along the boundary thus, we have two cycles α_i, β_i



on this fixed framed link we take

$$\alpha_i \longrightarrow -\beta$$
$$\beta_i \longrightarrow \alpha$$

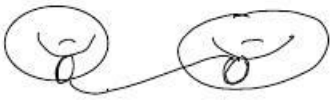
$\beta, \alpha \in H_1(\partial T_g; \mathbb{Z})$ which are standard longitude and latitude.

Thm: Any connected compact oriented 3-manifold M without

boundary can be obtained" as M_L for some framed link L in S^3.

Ex1: if $L =$  no twists then,

so, this is $L(0,-1)$ by our definition. we glue to solid tori with $\alpha \longmapsto -\alpha$

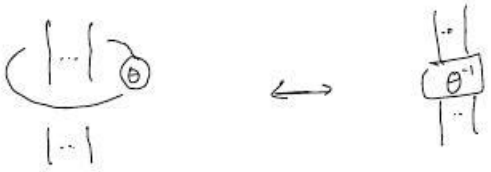


$\simeq S^2 \times S^1$

-  $\alpha \longmapsto \alpha+\beta$ $L(1,1) \simeq S^3$ similarly for 

Thm: [Kirby] $M_L \simeq M_{L'}$ iff L' can be obtained from L by Kirby moves,

① 

② 

} this is equivalent to the usual one!

Invarint of closed 3-manifolds

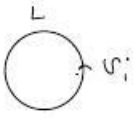
Now the we know that we can just describe any 3-manifold using a framed link L

so take

$L \xrightarrow[\text{in } M]{\text{color}} F(L, \Omega)$ ↑ evaluate

we get then an invarint if we color by an simple e $Inv(e)$

let us say for now that L has just one component, and we want to use all information in e. So, we color using all the sim

$\sum \lambda_i$  $\equiv F(L, \Omega)$

add another components we use the same coloring summing over all c
it's like multiplying

$(\sum \lambda_i \langle v_i \rangle)^2$ in $K(\mathcal{C})$ the Grothendieck ri

$\sum_{i,j} \lambda_i \lambda_j$  and so on!

So, basically

Def 1: let L be a link with m-components then,

$$\langle L, \Omega \rangle = \sum_{\lambda \in Col(L)} \left(\prod_{n=1}^{m} \lambda_n \right) F(L, \lambda)$$

That seems ok! But

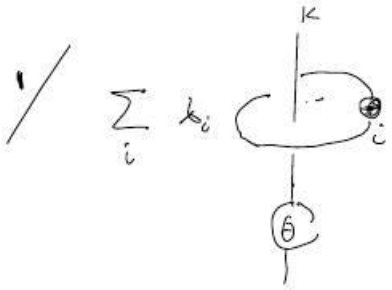
if we want this to be an invariant of 3-manifolds
$\langle L, \Omega \rangle$ must not change under Kirby moves!! Also
what are those λ_s?

we can see that these categories are invariant under permutation of simples!
So basically λ_is cannot be zero $\forall i$ also if they are all different they must be related the colorly λ_i in some way?

Prop: Kirby move (2) fixes $\lambda_i = d_i$ the dimension

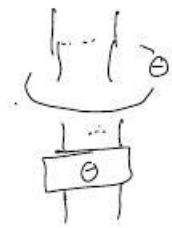
To see this we find some relations that are fixed by (2)
Since we can do everything locally, we just focus at the small

Rmk: one should note that the invariant doesnot depends on the choosen of L. ($v_i \leftrightarrow v_i^*$)

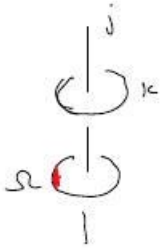
1/  $\longleftrightarrow$  $r=1$

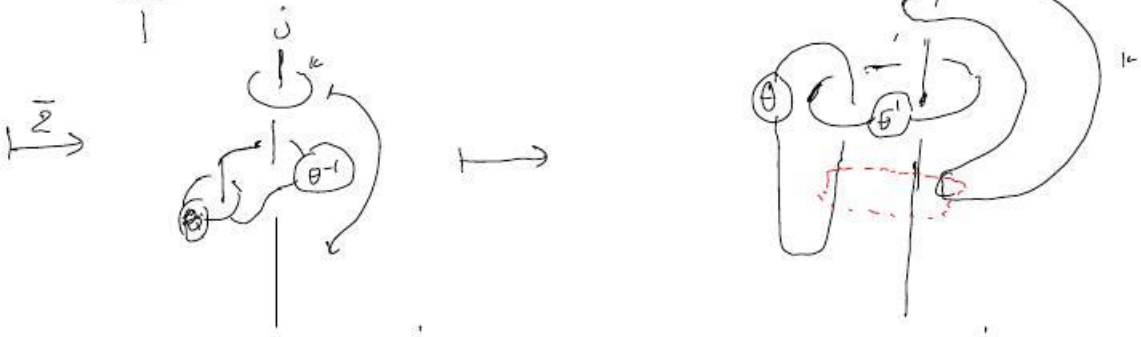
$$\Rightarrow \boxed{\sum_i \lambda_i \, \theta_i : \theta_{i^*} \quad \frac{S_{i\,i^*}}{d_{i^*}} = 1} \qquad \text{(requirement 1)}$$

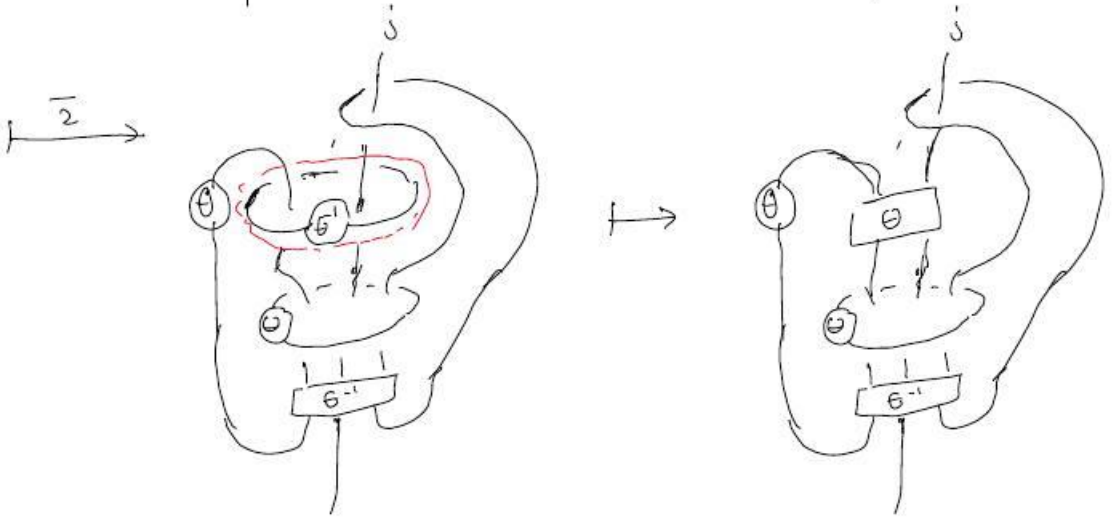
it's easy to see afterwards that

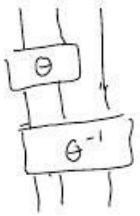
 $=$  just by  and applying the formulas!

Then we check the following

 $\sum_i \lambda_i \frac{S_{ij}}{d_j} \frac{S_{i^*j}}{d_j}$



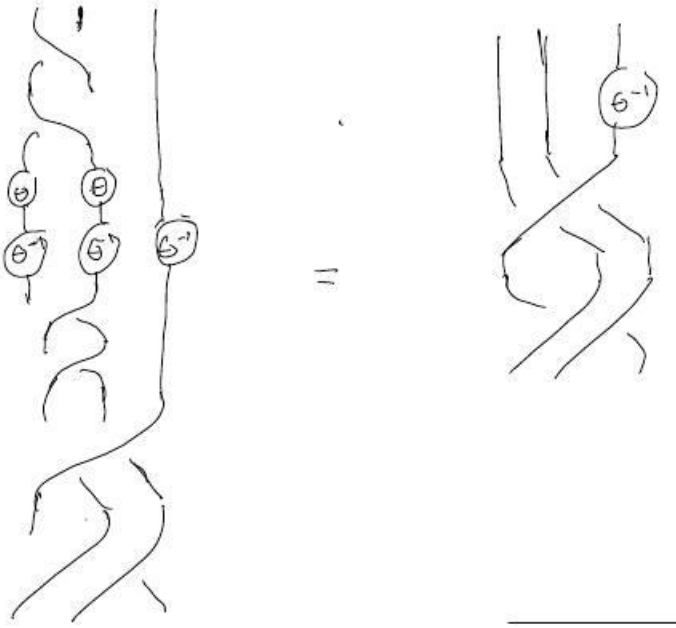


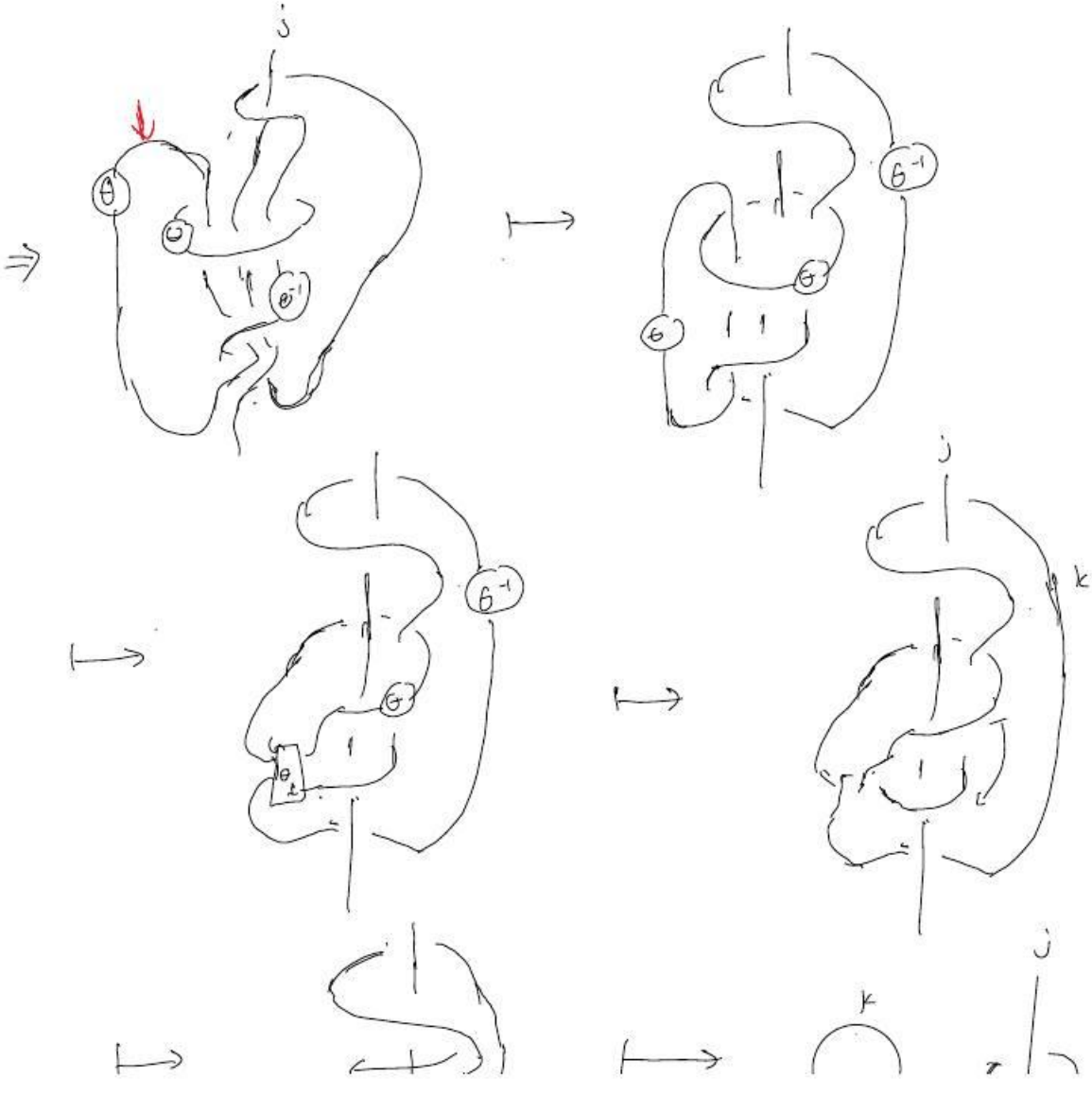


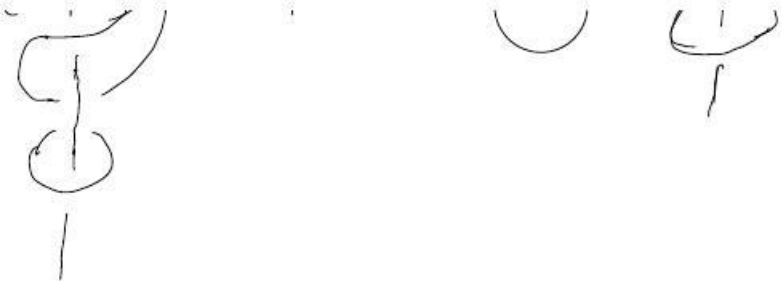
we know $\Theta_{X\otimes Y} = (\Theta_X \otimes \Theta_Y) \circ C_{Y,X} \circ C_{X,}$

$\Theta_{(X\otimes Y)\otimes Z} = (\Theta_{X\otimes Y} \otimes \Theta_Z) \circ C_{Z,X\otimes Y} \circ C_{X\otimes Y,}$

$= \Big(\big[(\Theta_X \otimes \Theta_Y) \circ C_{Y,X} \circ C_{X,Y} \big] \otimes \Theta_Z \Big) \circ C_{Z,X\otimes Y} \circ C$




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$$\Rightarrow \quad \sum_i \underbrace{\lambda_i \frac{S_{i,j}}{d_j}}_{\omega_j} \frac{S_{k,j}}{d_j} = d_k \sum_i \lambda_i \frac{S_{i,j}}{d_j}$$

$$\Rightarrow \quad \omega_j S_{k,j} = d_k d_j \omega_j$$

now we use the invertability of the S matrix

notice for $d_k^{-1} d_j^{-1} S_{k,j}$] $\tilde{S}$ has the first colum / row just ones



Since $(\tilde{S}^{-1} S)_{ij} = \delta_{i,j}$

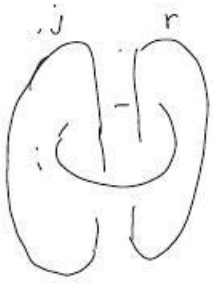
$$\Rightarrow \quad \sum_k \tilde{S}^{-1}_{i,k} S_{k,j} = \delta_{i,j}$$

$$1 = \sum_k \tilde{S}^{-1}_{0k} = \sum_k \tilde{S}^{-1}_{k0}$$

$$\Rightarrow \quad \omega_j = \sum_k S^{-1}_{k0} \omega_j = \sum_k \omega_j \tilde{S}^{-1}_{k,0} \tilde{S}_{k,j}$$

$$= \omega_0 \delta^0_j$$

$$\Rightarrow \quad \sum_i \lambda_i S_{0,i} = \omega_0$$

$\sum \lambda_i$ 

$$= \sum \lambda_i N^k_{j,r}$$ 

$$= \sum \lambda_i N^k_{j,r} \frac{S_{k,i}}{d_k}$$ but

$$= \sum N^0_{i,r}\ \lambda_i\ S_{0,i}$$

$$N^0_{i,r} = \dim \mathrm{Tr}\,(\ ,\ V_i \otimes V_r^*) = 1\,\delta_{i,r}$$

$$\Rightarrow \quad \boxed{\delta_{i,r} \sum_i \lambda_i d_i = \sum_i \lambda_i \frac{S_{i,i} S_{i,r}}{d_i}} \quad (\text{req } 2)$$

$$= x\,\delta^r_i$$

from requirment

$$\sum_i \lambda_i \theta_i \theta_j S_{i,j} = d_j$$

$$\sum_i \lambda_i \lambda_j \theta_i \theta_r^{-1} \frac{S_{j,r}}{d_j} S_{i,j} = \sum_j \theta_r^{-1} \theta_j^{-1} \lambda_j S_{j,r}$$

$$\Rightarrow \sum_i \theta_r^{-1} \theta_i \lambda_i \underbrace{\sum_j \lambda_j \frac{S_{i,j}}{d_j} S_{j,r}}_{\times\, \delta_{i,r}} = \sum_j \theta_r^{-1} \theta_j^{-1} \lambda_j S_{j,r}$$

$$\Rightarrow \quad \boxed{\lambda_r\, x = \sum_j \theta_r^{-1} \theta_j^{-1} \lambda_j S_{j,r}} \quad (\text{eq } 3)$$

Next, we calculate



$$\Rightarrow \quad x\, d_j = \sum \lambda_i \theta_j^{-1} \theta_i^{-1} S_{i,j}$$

$$= x \lambda_j \quad \sim \quad \boxed{d_j \propto \lambda_j}$$

Now we focus on

Kirby move 1

$L \sqcup \bigcirc^{\pm 1}$

$\Omega = \sum d_i V_i$

$$\langle L, \Omega \rangle = \sum_{\lambda \in \mathrm{col}(L)} \left(\prod d\right) F(L, \lambda)$$

$$\langle L \cup S^{\pm}, \Omega \rangle = \langle L, \Omega \rangle$$

each time we add $\bigcirc^{\pm}$ we get the following factor

$$\sum_{i=1}^{n} \theta_i^{\pm 1} d_i^2 \equiv P_{\pm}$$

how many circles you add affects the number of components of the whole link and multiplies by this factor

now, we need something to track this down! The exact place to look is the linking matrix of the L. each time we do this change our lin matrix M changes by a block diagonal ± 1.

$\begin{pmatrix} L & & \\ & \pm & \\ & & \ddots \pm \end{pmatrix}$ depending on how many components we added

$\Rightarrow$ $\sigma(M)$ the signature just get $\pm n$ values!

Cool! This just implies that we need to normalize by

$$\tilde{N} = x^{|L|} y^{\sigma(L)}$$

$$\Rightarrow \quad x^{|L|+1} y^{\sigma(L)+1} P_+ = x^{|L|} y^{\sigma(L)}$$

$$\Rightarrow \quad xy = \frac{1}{P_+}, \qquad \frac{x}{y} = \frac{1}{\Delta_-}$$

$$\Rightarrow \quad x^2 = \frac{1}{P_+ P_-} = \frac{1}{D^2}, \quad y = \frac{D}{P_+}$$

$$N = \left(D^{-1}\right)^{|L|} \left(\frac{\sqrt{P_+ P_-}}{P_+}\right)^{\sigma(L)}$$

$\cdots \quad \cdots \quad \cdots \quad \

$$= \mathcal{D}^{-|L|} \left(\frac{p_-}{p_+}\right)^{\sigma(L)/2} = \mathcal{D}^{-|L|} \left(\frac{p_+}{p_-}\right)$$

one last thing!

- They usually rewrite this by requiring further normalization

$$\tau(S^2 \times S^1) = \langle \bigcirc, \Omega \rangle$$

$$= \mathcal{D}^{-1} \underbrace{\sum d_i^2}_{\mathcal{D}^2} = \mathcal{D}$$

So, we normalize by $\mathcal{D}$

$$\tau(M) \equiv \langle L, \lambda \rangle = \mathcal{D}^{-|L|-1} \left(\frac{p_+}{p_-}\right)^{-b(L)/2} \sum_{\lambda \in Col(L)} \left(\prod d_i\right) F(L, \lambda)$$

Now, let us try to promote this invariant to a full TQFT meaning making it assign values to 3-manifold with boundary and further to surfaces!